

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2017
Department of Materials Science and Engineering

Ch 1227
Organic Chemistry

Time: 3 hours

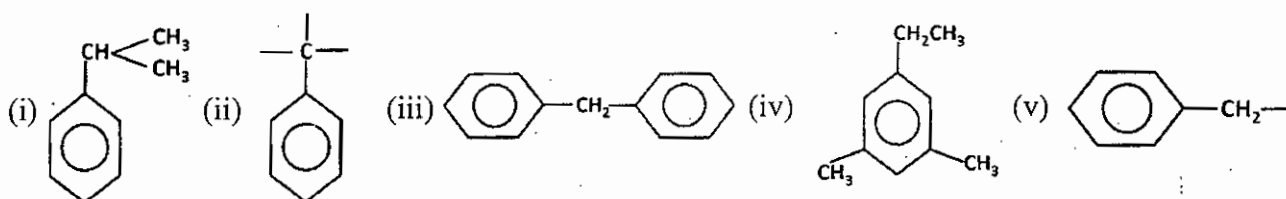
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What is hydrogen bonding? How does it modify the properties of organic compounds? (08)
(b) What are sigma bond and pi-bond? Explain why two S-orbitals do not form a pi-bond? (07)
(c) Write the differences between valence band theory and molecular orbital theory. (10)
(d) O₂ (oxygen) molecule is more stable than O₂⁻ (oxygen) ion, explain on the basis of molecular orbital theory. (10)
- Q 2. (a) Discuss the aromaticity of benzene with reference to (i) heat of hydrogenation or heat of combustion (ii) bond length. (10)
(b) Which of the following compounds will show aromaticity, Give reasons. (12)
(i) Cyclopentadienyl anion (ii) Pyrrole (iii) Pyridine
(c) Phenols are acidic in character, while alcohols are almost neutral. Explain in detail. (08)
(d) Write the name of the following compounds: (05)

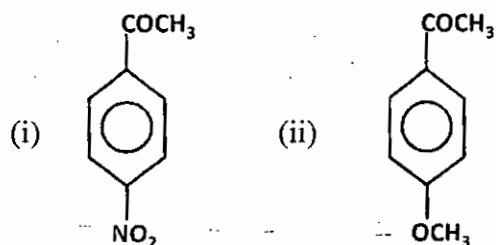


- Q 3. (a) Write the name of three main steps of Polymerization reaction and describe them briefly with suitable example. (10)
(b) How do thermoplastic polymers differ from thermosetting polymers? Give two examples of each type. (10)
(c) Define Isotactic and Atactic polymer with examples. (08)
(d) Calculate the number and weight average molecular weight of a polymer sample containing 15, 10 and 100 chains of molecular weight 1015 g/mol, 1200 g/mol and 1350 g/mol respectively. (07)
- Q 4. (a) How are aromatic amines defined? Illustrate with examples. (08)
(b) Pyrrole is weakly acidic whereas Pyridine is weakly basic-Explain. (10)
(c) CHCl₃ is polar but CCl₄ is nonpolar explain, why? (08)
(d) Arrange the following compound in order of their increasing bond angle and explain with reason: H₂O, H₂Te, H₂Se and H₂S. (09)

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Write down the basic principle of ultraviolet and infrared spectroscopy. (10)
(b) How will you distinguish the following compound with the help of IR spectroscopy (05)



- (c) Discuss graphically the possible electronic transition and give example of each transition. (10)
(d) Define the term "Chromophore". Graphically explain the shifts and effects of absorbance and wavelength based on chromophore. (10)
- Q 6. (a) S_N2 reaction competes with E2 reaction, in respect of nature of nucleophiles, size, and nature of substrate-Explain with examples. (10)
(b) What is Hoffman and Zaister rule? Explain with example. (07)
(c) S_N2 reaction proceeds with complete stereochemical inversion-Explain with example. (08)
(d) Why an S_N1 reaction proceeds with racemization? (10)
- Q 7. (a) Graphically explain the change of structure of amino acids, when plotting concentration verse pH of amino acids. (10)
(b) What are essential amino acids? Why α -amino acids (except glycine) are optically active? (07)
(c) Explain how peptides are form from amino acids. (09)
(d) Write short note on: (i) N-terminal residue (ii) Basic amino acids (iii) Denaturation. (09)
- Q 8. (a) Briefly discuss the structure and properties of cellulose. (08)
(b) What happen when cellulose react with concentrated nitric acid in presence of sulphuric acid? Give reaction. (09)
(c) A conjugated diene absorbs at higher wavelength compared to diene in which double bonds are isolated. Explain with example. (08)
(d) Discuss the primary structure of a Protein. What is the importance of the primary structure? (10)

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B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2017
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Ph 1227
Magnetism and Nuclear Physics

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Define the permeability, magnetic susceptibility and magnetic induction. Prove that $\mu_r = 1 + \chi$ and hence show that $B = \mu_0(1 + \chi)H$. (12)
- (b) What is meant by magnetic circuit? In what respects does a magnetic circuit differ from an electric circuit? (13)
- (c) An iron rod 85 cm long and 3 mm² cross-section is placed in a long solenoid of 38 turns cm⁻¹ carrying a current of 2.7 amp. The rod gains a magnetization of 14.5×10^5 A/m. Determine the magnetic moment, susceptibility and the magnetic field. (10)
- Q 2. (a) State Ampere's law. Apply Ampere's law to obtain an expression for the force per unit length between two parallel current carrying conductors. (13)
- (b) Explain the concept of dielectric and dielectric constant. Develop expression for dielectric constant with and without taking into account local fields. (12)
- (c) Calculate the electronic polarisability of liquid Xenon whose dielectric constant is 1.67 and density 3.1 gm/cm³. Find the dielectric constant using Clausius-Mossotti equation. Given the argon atoms per unit volume = 1.37×10^{28} /m³. (10)
- Q 3. (a) Explain the meaning of each Maxwell's equation. Derive Maxwell's four equations of electromagnetism in free space. (15)
- (b) Distinguish between self inductance and mutual inductance. Derive an expression for the energy stored in an inductor. (10)
- (c) An iron ring has a cross-sectional area 10 cm² and air gap of width $d = 2$ mm and mean length, $l = 2\pi r = 60$ cm including the air gap. Find the number of turns required to produce flux density $B = 1$ wb/m². Given, $\mu_{\max} = 10^{-3}$ H/m (10)
- Q 4. (a) What is meant by hysteresis loop? Explain coercivity and remanence. How does the study of hysteresis help us in the choice of magnetic materials? (10)
- (b) Derive an expression for paramagnetic susceptibility according to classical theory. Comment on the temperature dependence of susceptibility. (15)
- (c) An electron moves at a speed of 6×10^6 m/s perpendicular to a constant magnetic field. The path is a circle of radius 1.3×10^{-3} m. (i) What is the magnitude of the field? (10)
- (ii) Find the magnitude of the electron's acceleration.

Section B

(Answer ANY THREE questions from this section in **answer script B**)

- Q 5. (a) What is elasticity? Explain the general behavior of material under stress using a stress-strain curve. (10)
- (b) Define Young's modulus, rigidity modulus and Poisson's ratio and obtain the relation connecting the three. Give an idea about factor of safety. (15)
- (c) Calculate the work done in stretching a uniform metal wire of area of cross-section 10^{-6} m^2 and length $4 \times 10^{-3} \text{ m}$. Given, $Y = 2 \times 10^{11} \text{ Nm}^{-2}$ (10)
- Q 6. (a) What is a cantilever? Obtain an expression for the depression at the free end of a light beam clamped horizontally at one end and loaded at the other. (15)
- (b) Sketch the binding energy per nucleon versus mass number curve. What information can you obtain from this curve? (10)
- (c) Calculate the binding energy per nucleon of ${}^{64}_{28}\text{Ni} = 63.9280 \text{ amu}$. Mass of hydrogen atom (M_H) = 1.007825 amu, mass of neutron (M_n) = 1.008665 amu, mass of electron (M_e) = 0.00055 amu. (10)
- Q 7. (a) What is nuclear fusion? Discuss the underlying problem of nuclear fusion. (10)
- (b) Describe briefly the various components of a nuclear reactor. Explain the principle of its working. (15)
- (c) A city requires 100 MeV of electrical power on an average. If this is to be supplied by a nuclear reactor of efficiency 20% using U-235 as the nuclear fuel, calculate the amount of fuel required for one day's operation. Given that energy released per fusion of U-235 nuclide = 200 MeV. (10)
- Q 8. (a) Define activity as applied to the process of radioactive decay. Define main units of measuring intensity of radioactivity. (10)
- (b) Discuss the theory of successive disintegration of radioactive substance and obtain the conditions for transient and secular equilibrium. (15)
- (c) 1 gram of radioactive substance disintegrates at the rate of 3.7×10^{10} disintegrations per second. The atomic weight of the substance is 226. Calculate its mean life. (10)

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Math 1227
Differential Equation and Transform Analysis

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

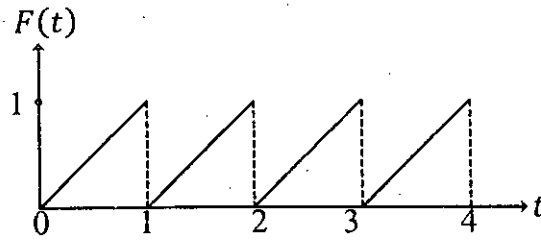
- Q 1. (a) Form differential equation for $xy = Ae^x + Be^{-x} + x^2$; where, A and B are parameters. Write down order and degree of your final differential equation. (12)
- (b) Solve: $x^2 \frac{dy}{dx} - 2xy = 3y^4; y(1) = \frac{1}{2}$ (10)
- (c) Solve: $\frac{dy}{dx} = \frac{x+3y+4}{2x+y+3}$ (13)
- Q 2. (a) State the specific reason(s) for what the following differential equations are nonlinear: $(1 - y)y' + 2y = e^x$; $\frac{d^2y}{dx^2} + \sin y = 0$; (05)
and $y''' + y^2 = \ln t$
- (b) Solve: (i) $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$ (07)
- (ii) $\sin x \frac{dy}{dx} + (\cos x)y = 0$ subject to, $y\left(\frac{7\pi}{6}\right) = -2$ (08)
- (c) Solve by variation of parameters: $y''' + y' = \tan x$ (15)
- Q 3. (a) Find the general solution of: $y'' - 5y' + 4y = 8e^x + \sin x$ (10)
- (b) Solve: $(D^2 - 9D + 18)y = e^{-3x}$ (12)
- (c) Develop the Fourier series for $f(x) = x \sin x; 0 < x < 2\pi$ (13)
and $f(x + 2\pi) = f(x)$
- Q 4. (a) Show that an odd function has no **cosine** terms in its Fourier series. (11)
- (b) Expand $f(x) = x^2, 0 < x < 2\pi$ in Fourier series. (12)
- (c) Find the half-range **sine** series for the function f , defined by (12)

$$f(x) = \begin{cases} x & \text{for } 0 \leq x < \pi/2 \\ \pi - x & \text{for } \pi/2 \leq x \leq \pi \end{cases}$$

Section B

(Answer ANY THREE questions from this section in **answer script B**)

Q 5. (a) Find the Laplace transform of $F(t)$. Where, $F(t)$ is defined graphically below (11)



(b) Using convolution theorem, evaluate: $L^{-1} \left\{ \frac{1}{s^2(s^2 + 4)} \right\}$ (10)

(c) Solve: $y' + y = f(t)$, $y(0) = 5$ Where, $f(t) = \begin{cases} 0, & 0 \leq t < \pi \\ 3 \sin t, & t \geq \pi \end{cases}$ (14)

Using Laplace Transformations:

Q 6. (a) Express $f(x) = x^4 + 3x^3$ in terms of Legendre Polynomial. (12)

(b) Prove that, $\int_{-1}^1 P_m(x)P_n(x)dx = 0$, when $m \neq n$ (13)

(c) Show that, $\int_{-1}^1 xP_n(x)P_{n-1}(x)dx = \frac{2n}{4n^2-1}$ (10)

Q 7. (a) Define ordinary and singular point of a differential equation with example. (05)

(b) Use the method of Forbenius to find the solution of the differential equation (18)

$$2x^2y'' - xy' + (x - 5)y = 0, \text{ about } x = 0$$

(c) Evaluate: $L^{-1} \left\{ \frac{5s + 3}{(s-1)(s^2 + 2s + 5)} \right\}$ (12)

Q 8. (a) Prove that, $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ (09)

(b) Prove that, $x J'_n = n J_n - x J_{n+1}$ (12)

(c) Prove that, $e^{\frac{1}{2}x(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} t^n J_n(x)$ (14)

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B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2017
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ME 1227
Engineering Mechanics

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What is meant by Engineering Mechanics? Explain the parallelogram law for the addition of forces. (05)
- (b) The five forces P, Q, R, S and T act on a bolt A as shown in fig. 1(b). Determine the resultant of the forces on the bolt. (See fig. 1(b) on page 3) (13)
- (c) A transmission tower is held by three guy wires anchored by bolts at B, C and D. If the tension in wire AB is 2100 N, determine the components of the force exerted by the wire on the bolt at B. (See fig. 1(c) on page 3) (17)
- Q 2. (a) What is meant by couple? The magnitude and direction of two forces are $P = +12\text{ N}$ and $Q = -13\text{ N}$. Are the forces P and Q forms a couple or not? Explain. (05)
- (b) Gear C is rigidly attached to Arm AB. If the forces and couple shown in fig. 2(b), can be reduced to a single equivalent force at A, determine the equivalent force and the magnitude of the couple M. (See fig. 2(b) on page 3) (17)
- (c) The connecting rod AB exerts on the crank BC a 1.5 KN force directed down and to the left along the center line of AB. Determine the moment of the force about C. (See fig. 2(c) on page 3) (13)
- Q 3. (a) Define and explain free body diagram. (05)
- (b) A 2.4 m boom is held by a ball and socket joint at C and by two cables AD and AE. Determine the tension in each cable and the reaction at C. (See fig. 3(b) on page 3) (30)
- Q 4. (a) The coefficient of friction are $\mu_s = 0.40$ and $\mu_k = 0.30$ between all surfaces of contact. Determine the smallest force P required to start the 30 kg block moving if cable AB (i) is attached as shown (ii) is removed. (See fig. 4(a) on page 3) (18)
- (b) A 400 N weight is attached at A to the lever as shown in fig. 4(b). The constant of the spring BC is $k = 12\text{ KN/m}$, and the spring is unstretched when $\theta = 0$. Determine the position of equilibrium. (See fig. 4(b) on page 3) (17)

Section B

(Answer ANY THREE questions from this section in **answer script B**)

- Q 5. (a) Block B starts from rest and moves downward with a constant acceleration. Knowing that after slider block A has moved 400 mm its velocity is 4 m/s, determine (i) the acceleration of A and B, (ii) the velocity and the change in position of B after 2s. (See fig. 5(a) on page 4) (18)
- (b) A projectile is fired from the edge of a 150 m cliff with an initial velocity of 180m/s, at an angle of 30° with the horizontal. Neglecting air resistance, find (i) the horizontal distance from the gun to the point where the projectile strikes the ground, (ii) the greatest elevation above the ground reached by the projectile. (See fig. 5(b) on page 4) (17)
- Q 6. (a) Two wires AC and BC are tied at C to a 7 Kg sphere which revolves at a constant speed v in the horizontal circle shown in fig. 6(a). Knowing that $\theta_1=55^\circ$ and $\theta_2=30^\circ$ and that $d=1.4$ m, determine the range of values of v for which both wires remain taut. (See fig. 6(a) on page 4) (18)
- (b) A small sphere of mass 3 Kg is held as shown in fig. 6(b), by two wires AB and CD. If wire CD is cut, determine the tension in the other wire (i) before CD is cut, and (ii) immediately after CD has been cut. (See fig. 6(b) on page 4) (17)
- Q 7. (a) A 1.2 kg collar C may slide without friction along a horizontal rod. It is attached to three springs, each of constant $k=400$ N/m and 1500 mm undeformed length. Knowing that the collar is released from rest in the position shown in fig. 7(a), determine the maximum speed it will reach in the ensuing motion. (See fig. 7(a) on page 4) (17)
- (b) A 450 Kg pellet is released from rest at A and slides without friction along the surface as shown in fig. 7(b). Determine its velocity (i) as it passes through B, and (ii) as it passes through C. (See fig. 7(b) on page 4) (18)
- Q 8. (a) The velocity of two balls before the impact are as shown in fig. 8(a). If after the impact the velocity of ball B is observed to be 2.5 m/s to the right, determine the coefficient of restitution between the two blocks. (See fig. 8(a) on page 4) (17)
- (b) A pulley and two loads are connected by inextensible cords as shown in fig. 8(b). Load A has a constant acceleration of 300 mm/s^2 and as initial velocity of 240 mm/s, both directed upward. Determine (i) the number of revolutions executed by the pulley in 3s, (ii) the velocity and position of load B after 3s, (iii) the acceleration of point D on the rim of the pulley at $t=0$. (See fig. 8(b) on page 4) (18)

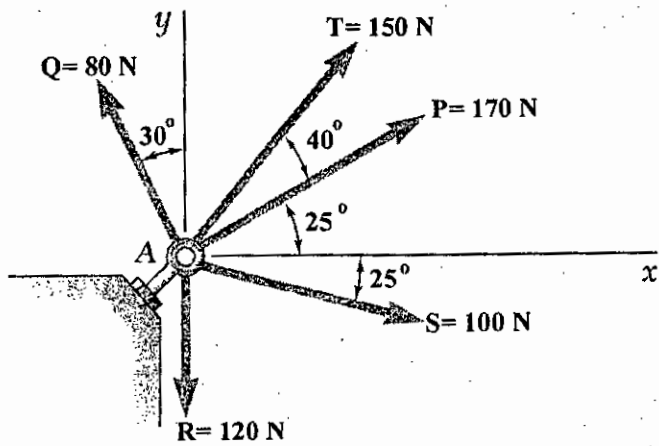


Fig.: 1(b)

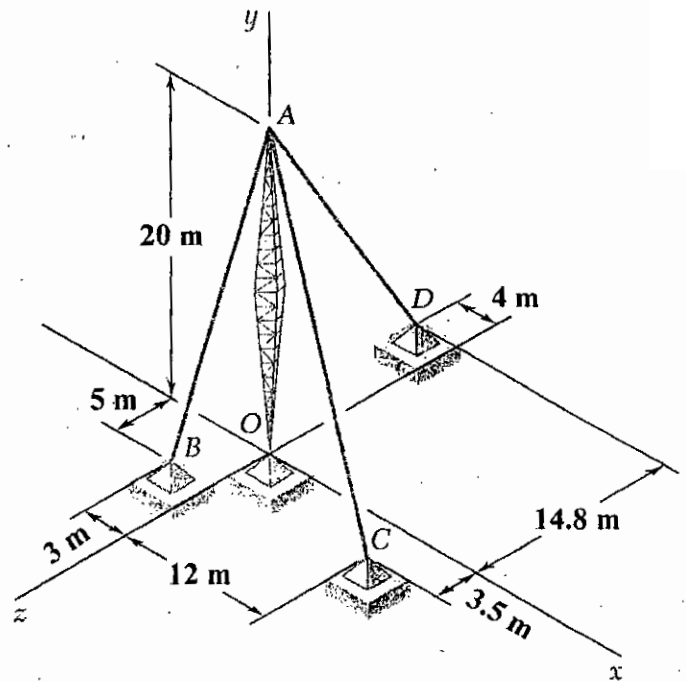


Fig. 1(c)

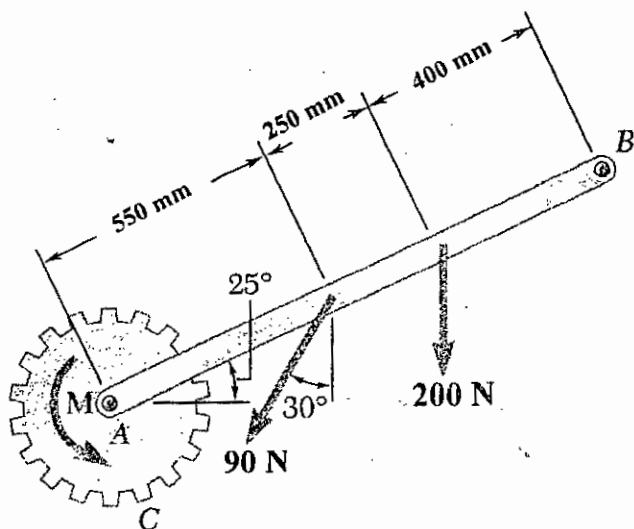


Fig. 2(b)

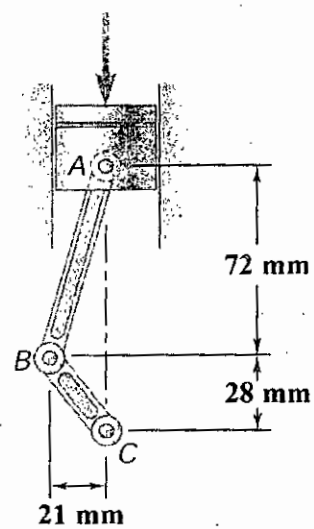


Fig. 2(c)

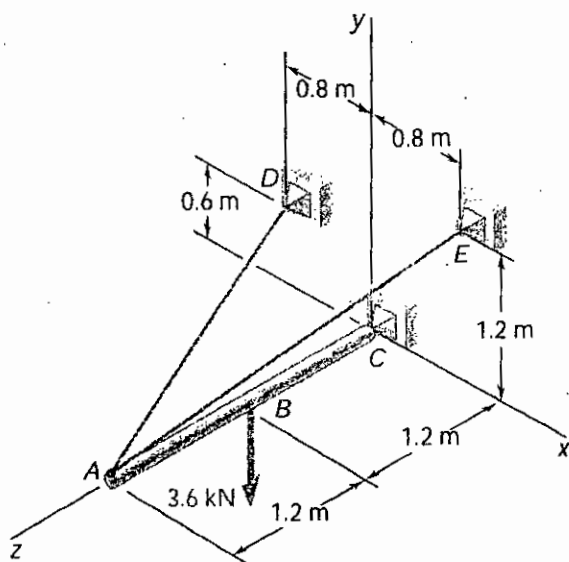


Fig. 3(b)

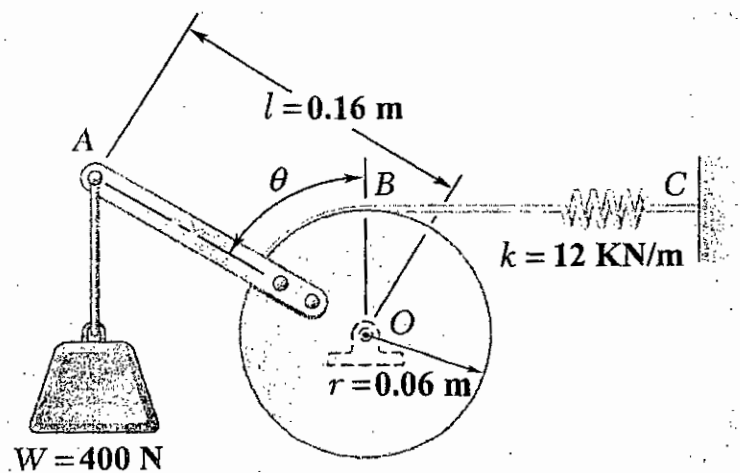


Fig. 4(b)

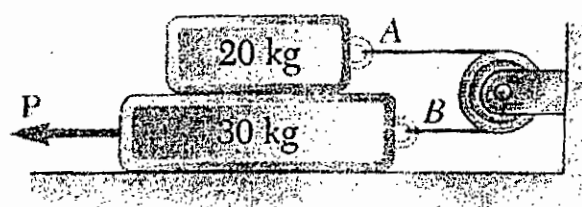


Fig. 4(a)

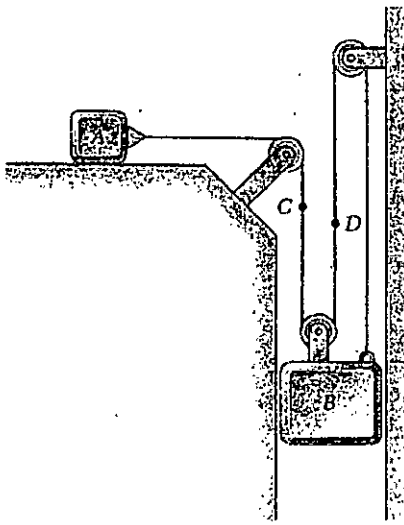


Fig. : 5(a)

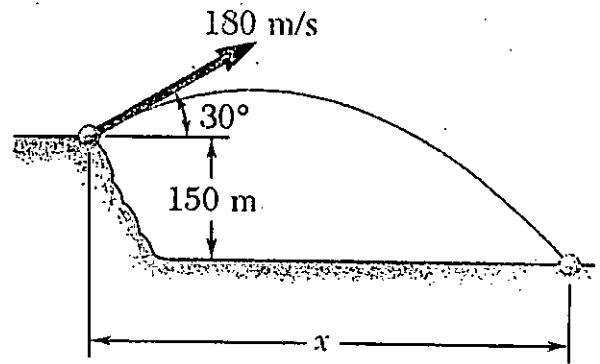


Fig. : 5(b)

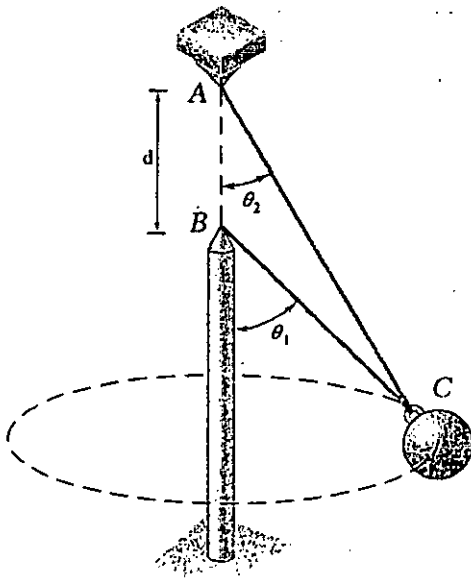


Fig. : 6(a)

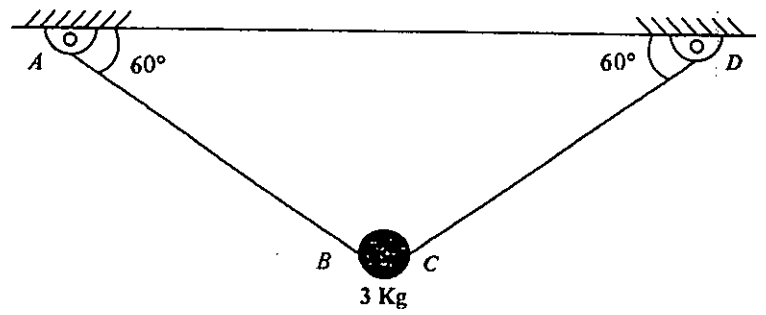


Fig. : 6(b)

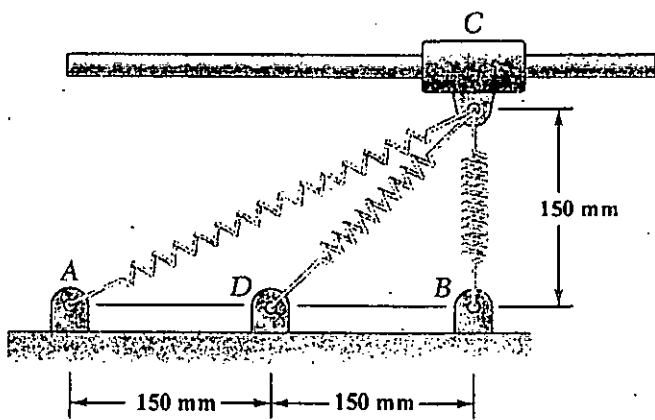


Fig. : 7(a)

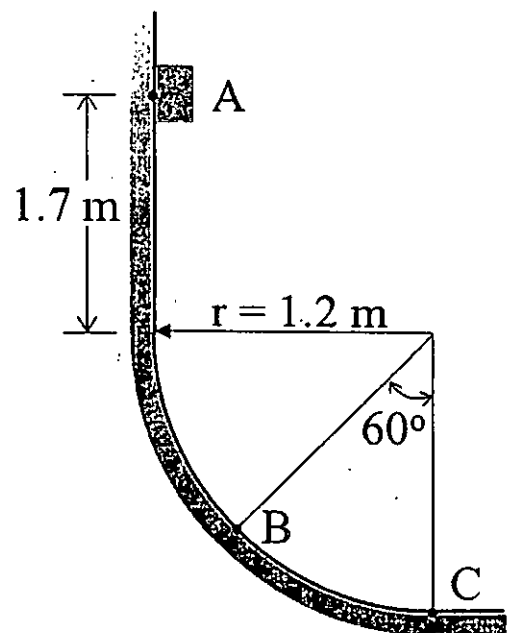


Fig. : 7(b)

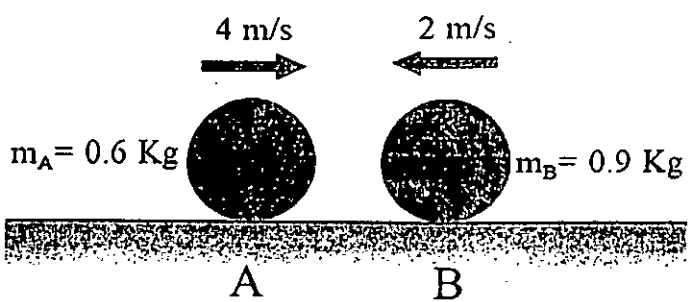


Fig. : 8(a)

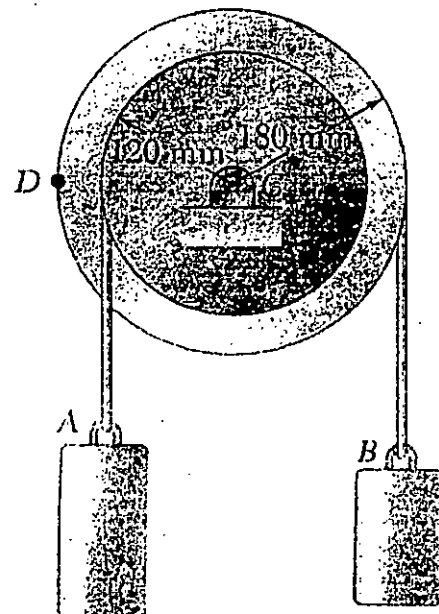


Fig. : 8(b)

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B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2017
Department of Materials Science and Engineering
Hum 1227
Technical English

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Make sentences with the following structures using the words given in brackets. (14)
- (i) Subject + Verb + Object + Adverb of manner (pollute as verb)
 - (ii) Subj. + Modifier + Verb + that + Subj. + Verb + Adv. of place (is and study as verb)
 - (iii) Subj. + Appositive + Linking verb + Adj. complement (seem as verb)
 - (iv) What + Subj. + Verb + Adv. of manner + Verb + Adj. complement. (demand and is as verb)
 - (v) Before + Subj. + Verb + Adv. of place, subj. + Verb + Adv. of place (came and were as verb)
 - (vi) Neither + Subj. + nor + Subj. + Verb + Object (obey as verb)
 - (vii) Subj. + relative pronoun + Verb + Object + Verb + Noun complement (demand and are as verb).
- (b) Make sentences using the following modals as directed. (12)
- (i) May (To express guess about the future)
 - (ii) Could (To express guess inference)
 - (iii) Should (To express duty in the past)
 - (iv) Must (To express logical deduction in the present)
 - (v) Need (To express unnecessary action in the past)
 - (vi) Had better (To express preference)
- (c) Make new words with the following prefixes and make sentence with the new words. (09)
- Ambi____, Bi____, Dia____, Em____, Im____, Sub____
- Q 2. (a) Make W.H. question with each of the underlined word/words of the following sentences. (14)
- (i) The plane flagrantly violated a UN no-fly zone.
 - (ii) The medical prognosis was bleak.
 - (iii) Soldiers trained to fight.
 - (iv) It's been lovely having you here.
 - (v) Her comment was a highly perceptive.
 - (vi) My father used to perch me on the front of his bike.
 - (vii) He finally received a cordial greeting.
- (b) Make sentences using the following words as directed: What (as pronoun); Still (as verb); Book (as verb); Book (as adjective); Near (as adverb) (12)
- (c) Write two synonyms for each of the following words and make sentence with the synonyms: (09)
- Lovely, Rude, Patience.
- Q 3. (a) Transform the following sentences as asked in the brackets (14)
- (i) Their love grew with the years. (Compound)
 - (ii) She, a symbol of social conscience, involves in politics. (Complex).
 - (iii) It looks a flagrant abuse of human rights. (Negative)
 - (iv) She not only answered all of my questions but also kept her honesty in them. (Simple)
 - (v) He possesses as pleasant in manners as his sister. (Comparative)
 - (vi) He, a sensible person, has always been good at managing his family. (complex)
 - (vii) He leads a good life, so is popular. (Simple)
- (b) Correct the following sentences : (12)
- (i) I don't mind to have an ice-cream.
 - (ii) Tell me what is your name.
 - (iii) You and he should do his duty.
 - (iv) Many a rich men live there.
 - (v) I saw the car of Hasan which is costlier than yours.
 - (vi) They should refrain to do it.
- (c) Define transitive verb, intransitive verb and infinitive with example. (09)

- Q 4. (a) Complete the sentences with subordinate clause as directed. (14)
- (i) I hope..... (noun clause)
 - (ii)is true. (noun clause)
 - (iii) Democracy is a notion..... (adjective clause)
 - (iv) The man.....is not in his office. (adjective clause)
 - (v) They were order to wait..... (adverb clause of time)
 - (vi) Read it very carefully..... (adverb clause of purpose)
 - (vii), I would have gone there. (adverb clause of condition)
- (b) Express the following notions / function in sentence: (12)
- (i) Invitation (ii) Dare (iii) Hatred (iv) Patience (v) Desertion (vi) Assurance
- (c) Supply a suitable word to fill in the gaps. (09)
- (i) They are on a picnic.
 - (ii) What.....you do on holidays?
 - (iii) He is fond of.....cricket.
 - (iv) He took.....his coat when he entered his house.
 - (v) Which one of the pensyours?
 - (vi) The ladder on I was standing was about to fall.

Section B

(Answer **ANY THREE** questions from this section in answer script B)

- Q 5. (a) Read the passage and answer the following the questions: (20)

Pasteur found that many diseases are due to tiny living creatures and germs, which dwell in and feed on living bodies and poison them. There are millions of germs living in our body, in the clothes we wear, and in the food we eat. If we are well and strong, they do not usually do us any harm; but if we are tired or run-down, the germs can attack us and make us ill. In particular, if we cut ourselves, the germs can make their way into the cut and poison us. Now, Pasteur's discovery was important because it taught the doctors that one of the best ways of caring disease was to find out the nature of the attacking germs that caused the disease, and then to kill them or help the body to kill them. Even more important, it taught them to be clean. When a surgeon operates today, he takes immense trouble to wash his hands and the instruments he is going to use, and not only he washes them but also boils them. This is called sterilizing them, and the object is to kill the germs with which the instruments are infected; for dirt of any kind is a great germ carrier.

Questions:

- (i) What did Pasteur discover about the causes of many diseases?
- (ii) What does make the body an easy prey to such diseases?
- (iii) What is the importance of Pasteur's discovery in the prevention and treatment of infectious diseases?
- (iv) How has it helped surgery?

- (b) Make a précis of the above passage (Q5.a) with a title. (15)

- Q 6. (a) Write a letter to the editor of a newspaper commenting on campus violence and its impact. (20)

- (b) Write a report on the sports day you recently observed at your varsity. (Around 800 words). (15)

- Q 7. (a) Amplify the idea 'Diligence is the mother of good luck.' (Around 1200 words) (20)

- (b) Write a contrast paragraph between public university and private university. (15)

- Q 8. Write a free composition on one of the followings: (35)

- (a) "Independence Day"
- (b) "Hard labor and fortune"